

Analytical Problem Solving1[← Back](#)**QUESTIONS****Q1** ✓
Prime Number
10 marks**Q2** ✓
Fibonacci series
10 marks**Q3** ✓
String
10 marks**Q4** ✓
ODD or EVEN
10 marks**Q5** ✓
Print numbers from 1 to 10
10 marks

5/5 answered

Q1 Prime Number

10 marks

A mathematical analysis system needs to verify whether a given number is prime. A prime number is a number greater than 1 that has no divisors other than 1 and itself.

Write a Python program that:

- Accepts an integer input from the user
- Checks whether the number is prime using appropriate control flow constructs
- Displays a suitable message indicating whether the number is prime or not

Ensure proper use of loops and conditional statements.

Your Code

```
1 n=int(input())
2 if n<=1:
3     print("Not a Prime number")
4 else:
5     for i in range(2,n):
6         if n%i==0:
7             print("Not a Prime number")
8             break
9         else:
10            print("Prime number")
11            break
```

Input

7

Output

Prime number

✓ Submitted



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QUESTIONS

Q1

Prime Number
10 marks



Q2

Fibonacci series
10 marks



Q3

String
10 marks



Q4

ODD or EVEN
10 marks



Q5

Print numbers from 1 to 10
10 marks



5/5 answered

Q5 Print numbers from 1 to 10

10 marks

A program is intended to display numbers from 1 to 10 using a loop. However, some parts of the code are missing.

Complete the program by filling in the blanks so that it correctly prints numbers from 1 to 10.

Program to print numbers from 1 to 10

for i in _____:

print(_____)

Also explain how the range() function works in this context.

Your Code

```
1 for i in range(1,11):  
2     print(i,end=" ")
```

Input

stdin input

Output

```
12345678910
```



Submitted



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Q4

ODD or EVEN
10 marks



Q5

Print numbers from 1 to 10
10 marks



5/5 answered

Q4 ODD or EVEN

10 marks

A program is designed to check whether a number entered by the user is even or odd. However, the program contains syntax and logical errors.

Analyze the given code and perform the following:

- Identify all errors present in the program
- Rewrite the corrected program
- Briefly justify each correction

```
num = input("Enter number: ")  
if num % 2 == 0  
print("Even")  
else  
print("Odd")
```

Your Code

```
1 num=int(input())  
2 if num%2==0:  
3     print("Even")  
4 else:  
5     print("Odd")
```

Input

4

Output

Even



Submitted



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Q5

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5/5 answered

Q3 String

10 marks

A text processing system receives user input in the form of a string that may contain alphabets, digits, and special characters.

Write a Python program that:

- Accepts a string input from the user
- Counts the number of vowels, consonants, digits, and special characters present in the string
- Displays the count for each category

Use appropriate string operations and control flow statements.

Your Code

```
1 s=input()  
2 v=c=d=sp=0  
3 for ch in s:  
4     if ch.isalpha():  
5         if ch.lower() in 'aeiou':  
6             v+=1  
7         else:  
8             c+=1  
9     elif ch.isdigit():  
10        d+=1  
11    else:  
12        sp+=1  
13 print("vowels:",v)  
14 print("Consonants:",c)  
15 print("Digits:",d)  
16 print("Special characters:",sp)
```

Input

Hello123@&

Output

```
vowels: 2  
Consonants: 3  
Digits: 3  
Special characters: 2
```



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String
10 marks



Q4

ODD or EVEN
10 marks



Q5

Print numbers from 1 to 10
10 marks



5/5 answered

Q2 Fibonacci series

10 marks

A numerical computation module is designed to generate a sequence where each number is the sum of the two preceding numbers. This sequence is known as the Fibonacci series.

Write a Python program that:

- Accepts a number n from the user
- Generates and displays the first n terms of the Fibonacci sequence
- Uses looping constructs to compute the sequence

Ensure the program handles cases where n is less than or equal to zero.

Your Code

```
1 n=int(input())
2 a=0
3 b=1
4 for i in range(n):
5     print(a,end=" ")
6     c=a+b
7     a=b
8     b=c
```

Input

7

Output

0112358



Submitted